

Time Series & Forecasting – Practical Examination Questions

1. Using any dataset from the Time Series Data Library (TSDL), plot the time series and identify the presence of trend, seasonality, and irregular components. Also draw the ACF and PACF plots and interpret the nature of the series.
2. Given a monthly sales dataset, apply a 3-point and 5-point Moving Average. Then perform Simple Exponential Smoothing (SES) using $\alpha = 0.3$. Compare the outputs and comment on how smoothing affects the series.
3. Select any time series (e.g., stock prices, rainfall). Perform a stationarity check using the Augmented Dickey-Fuller (ADF) test. Plot the rolling mean and rolling standard deviation. Convert the series into a stationary series using differencing and show the before/after results.
4. Using Python or Excel, decompose a time series into its Trend, Seasonality, and Residual components using additive decomposition. Explain whether additive or multiplicative decomposition is more suitable for the chosen dataset and justify your answer.
5. Simulate an AR(1) model defined as $X_t = 0.6X_{t-1} + \varepsilon_t$ and an MA(1) model. Plot both simulated series and explain the difference in behavior between AR and MA processes based on the graphs.
6. Using quarterly production data, identify suitable values of p, d, and q using ACF and PACF plots. Fit an ARIMA(p, d, q) model and report the AIC value. Justify why this model is appropriate for the dataset.
7. Take a seasonal dataset (e.g., monthly airline passenger data). Determine the seasonal parameters P, D, Q, and m. Fit a SARIMA model and forecast the next 12 time periods. Provide the forecast plot along with interpretation.
8. Given four years of monthly product sales data with seasonality, perform forecasting using Excel by applying:
 - (a) Moving Average method
 - (b) Holt's Double Exponential Smoothing
 - (c) Holt-Winters Triple Exponential SmoothingForecast the sales for the 5th year.
9. Using any dataset from the Time Series Data Library, perform a complete ARIMA modeling pipeline including:
 - (a) Time series visualization
 - (b) Stationarity testing

- (c) Differencing
- (d) ACF/PACF analysis
- (e) ARIMA model fitting
- (f) Forecasting for the next 10 data points

Submit all graphs and results.

10. Using the same dataset, fit the following forecasting models:

- (a) Simple Exponential Smoothing
- (b) Holt's Method
- (c) Holt-Winters Method
- (d) ARIMA

Calculate MAPE or RMSE for each method and identify which model performs best with justification.